

400-Gbit/s Net Transmission Over a 56-m Single-Carrier and Single-Aperture Dual-Polarization Sub-THz Link

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Abstract We demonstrate a single-carrier and dual-polarization wireless point-to-point transmission at 287 GHz over 56 m with a net data rate of 400 Gbit/s using a single aperture. We employ a novel mHEMT receiver and PCS-16-QAM at 64 GBd, compliant with IEEE 802.15.3d. © 2026 The Authors

Introduction

Ultra-high data rate wireless communications is considered a key enabler for emerging applications supported by sixth-generation (6G) mobile networks, including autonomous systems, extended reality, and massive Internet of Things [1, 2]. In mobile fronthaul scenarios, high-speed wireless sub-THz links can complement mmWave and fiber-based solutions by delivering high data rates without requiring costly fiber deployment. Since the mmWave spectrum is already heavily regulated and widely utilized, the IEEE standardized the 252.72...321.84 GHz band (IEEE 802.15.3d) as a spectrum resource for sub-THz communications [1]. However, operation at such high carrier frequencies remains challenging due to severe free-space path loss and the limited availability of broadband low-noise transmitter and receiver hardware. Consequently, high antenna gains are required, resulting in narrow beams that favor fixed point-to-point links over distances of tens to hundreds of meters [3]. Early experimental demonstrations established the feasibility of transmission at 100 Gbit/s over 20 m at a carrier frequency of 237.5 GHz [4]. Subsequent optoelectronic systems increased line rates to 160 Gbit/s near carrier frequencies of 300 GHz over laboratory link distances [5]. More recent milestones in the 300 GHz frequency range include single-aperture, single-polarization line rates of 200 Gbit/s over 52 m [6], 232 Gbit/s over 200 m [7], and 250 Gbit/s over 55 m using photonic microcomb sources [8]. Further advances demonstrate 288 Gbit/s over 100 m using four subcarriers [9], as well as up to 374.4 Gbit/s over a 1 m, single-polarization, two-channel sub-THz link [10]. Considering a 20% FEC overhead, these results correspond to data rates of up to 240 Gbit/s. To further improve the data rate, the signal-to-noise ratio (SNR) can be optimized via improved transceiver circuits or higher antenna gains, and spectral efficiency can be maximized by advanced modula-

tion schemes like probabilistic constellation shaping (PCS). Additionally, while the bandwidth can be expanded across the broader sub-THz spectrum from 110 GHz to 500 GHz in laboratory experiments [11], practical systems must adhere to spectrum regulations, such as the IEEE 802.15.3d standard for the 300 GHz band. Beyond these SNR and bandwidth enhancements, data rates can be further scaled through polarization- and/or spatial-division multiplexing over a single [12] or multiple apertures [9, 11]. While polarization multiplexing can double the data rate using orthogonal polarizations, spatial multiplexing of parallel free-space beams either requires very large aperture separation or results in non-orthogonal partially overlapping beams [9], for which crosstalk cannot be fully compensated by multiple-input multiple-output (MIMO) equalization. Accordingly, aggregate data rates beyond 1 Tbit/s are only achieved over multiple apertures [9, 11] at the expense of significant hardware complexity, rather than within a single-aperture configuration compliant with spectrum regulations.

In this work, we demonstrate a 56-m single-aperture dual-polarization point-to-point wireless link operating at 287 GHz. The link transmits a data rate of 400 Gbit/s using a 64 GBd symbol rate, while fully complying with the IEEE 802.15.3d frequency band. This net data rate has been achieved by applying FEC decoding with a realistic spatially-coupled low-density parity-check (SC-LDPC) code. The intradyne experimental setup demonstrates a single-aperture sub-THz wireless link, where the full spectrum of both polarizations is simultaneously transmitted and received. This is achieved using a high-bandwidth dual-polarization optoelectronic transmitter, two novel mHEMT high-bandwidth all-electronic receivers combined with high-bandwidth orthomode transducers (OMTs), high-gain cascaded lens antennas, and MIMO digital signal processing. To the best of the authors' knowledge, this result represents one of

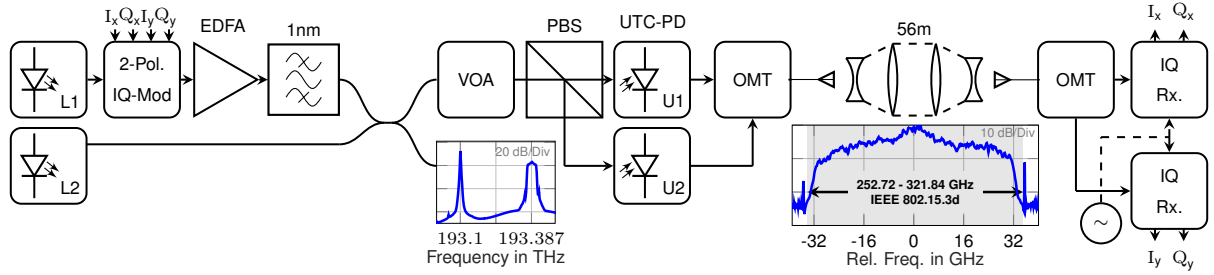


Fig. 1: Experimental setup of the dual-polarization point-to-point sub-THz transmission system. The optoelectronic transmitter (left) generates the 287 GHz carrier using optical heterodyne downconversion and dual-polarization IQ modulation, while the all-electronic coherent receiver (right) performs dual-polarization down-conversion and detection. Insets: Measured optical spectrum; baseband spectrum with the IEEE 802.15.3d band around 287 GHz as a reference.

the highest experimentally demonstrated data rate reported to date for a single-aperture and single-channel sub-THz point-to-point wireless link complying with the IEEE 802.15.3d band. Our excellent results establish this technology as a compelling solution for delivering fiber-class data rates in short-range, line-of-sight scenarios where rapid deployment, architectural flexibility, or physical constraints make the use of optical fiber impractical.

Wireless Transmission Setup

The point-to-point (P2P) sub-THz dual-polarization transmission system is based on a coherent dual-polarization optical transmitter combined with a dual-polarization all-electronic intradyne sub-THz receiver, as shown in Fig. 1. Dual-polarization baseband signals ($I_{x,y}$ and $Q_{x,y}$) are modulated onto an optical carrier generated by a laser (L1) using a LiNbO_3 dual-polarization IQ modulator. After amplification by an erbium-doped fiber amplifier (EDFA) and optical bandpass filtering, the modulated signal is combined with the output of a reference laser (L2) offset by 287 GHz using a 3 dB coupler. The combined optical signal is then split into two orthogonal polarizations by a polarization beam splitter (PBS) and individually detected by two broadband uni-travelling carrier photodiodes (UTC-PDs), thereby downconverting the optical signals to the sub-THz frequency range around 287 GHz. The resulting sub-THz signals, each corresponding to one polarization, are combined using a sub-THz OMT (OT034-MicroHarmonics) and radiated by a single dual-polarized circular horn antenna. The antenna's intrinsic gain of 26 dBi is

further enhanced by a cascaded concave-convex lens system, resulting in an overall gain of 51.1 dBi. This corresponds to a full-width-at-half-maximum divergence angle of 0.57° [13], making precise transmitter-receiver alignment essential, yet practically achievable. To ensure optimal beam coupling in our experiment, the transmit and receive antennas are manually aligned using three-axis rotation stages. At the receiver, the signal is captured by an identical cascaded lensed horn antenna. The two received polarizations are separated by a second OMT and processed by two synchronized intradyne, ultra-broadband, single-polarization, all-electronic receivers with integrated sub-harmonic mixers. These novel WR-3.4 waveguide receiver modules contain a fully integrated multifunctional front-end chip fabricated in Fraunhofer IAF's state-of-the-art 35-nm mHEMT technology. The inset in Fig. 1 shows a microscope image of the fully integrated mHEMT receiver chip. Each receiver circuit employs a fourfold electrical local-oscillator (LO) multiplication stage followed by an amplifier and a second-harmonic sub-THz IQ mixer, driven by a common 35.875-GHz LO. The incoming RF-signal is amplified by a three-stage low-noise amplifier (LNA) demonstrating a small-signal gain of 28 dB and a noise figure of approximately 7 dB over an operating bandwidth from 220 GHz to 330 GHz. As the transmit sequence, we use 25,600 random 16-QAM symbols, with and without PCS. The symbol rate is varied from 30 to 64 GBd. At the maximum symbol rate of 64 GBd, the transmitted spectrum, including a 5% spectral roll-off, spans from 253.4 GHz to 320.6 GHz and remaining entirely within the IEEE 802.15.3d band.

At the receiver, the electrical signals are acquired using a four-channel real-time oscilloscope with 80 GHz bandwidth and a sampling rate of 256 GSa/s. Offline digital signal processing (DSP) includes resampling to two samples per symbol, Godard timing recovery, periodogram-based frequency-offset compensation, 2×2 MIMO least-mean-square (LMS) adaptive equalization with 42 taps, and blind phase search (BPS) phase recovery with 3° -steps and a memory of 100 samples. After DSP, bit-wise log-likelihood ratios (LLRs) are

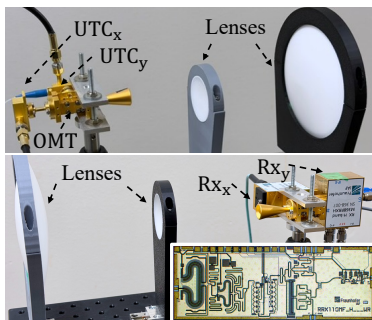


Fig. 2: Dual-polarization optoelectronic transmitter (upper) and electronic receiver (lower) with cascaded polytetrafluorethylene (PTFE) lens integrated conical horn antenna.

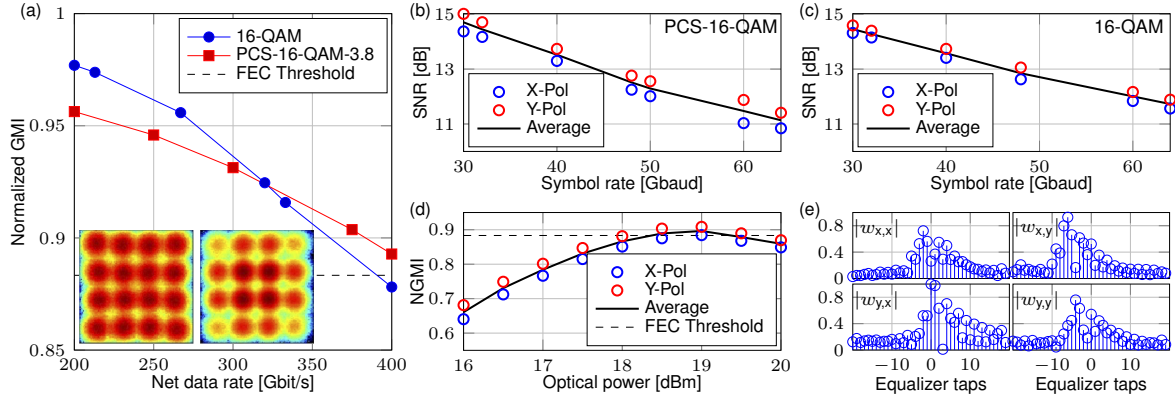


Fig. 3: Experimental results for 56-m point-to-point wireless transmission at 287 GHz. (a) Normalized generalized mutual information (NGMI) as a function of net data rate with 20% FEC overhead for 16-QAM and PCS-16-QAM, and constellation diagrams at 400 Gbit/s. Measured SNR over symbol rate for (b) PCS-16-QAM and (c) 16-QAM. (d) NGMI as function of Tx optical power for PCS-16-QAM at 64 GBd. (e) Magnitude of 2×2 MIMO equalizer taps for PCS-16-QAM at 64 GBd, running at 2 Sa./Sym.

then computed assuming an additive white Gaussian noise (AWGN) channel model [14], and the outcome is used to evaluate the SNR and the normalized generalized mutual information (NGMI) [15]. The transmission employs 16-QAM with optional PCS applied to enable flexible data-rate operation at a target entropy of approximately 3.8 bit/symbol.

Results

To evaluate the maximum transmission capacity, both 16-QAM and PCS-16-QAM signals are transmitted with increasing symbol rate, ranging from 30 GBd up to 64 GBd. The main experimental results are summarized in Fig. 3. Fig. 3(a) displays the polarization-averaged NGMI as a function of the net data rate, achieved by tuning the symbol rate, with and without PCS at the respective optimal transmit power. The net data rate is computed accounting for a 20% FEC overhead; for PCS, it is obtained assuming the use of probabilistic amplitude shaping (PAS) [16]. The NGMI threshold is set to 0.88, corresponding to a code rate of 0.83 with a 0.05 code gap [15]. For lower data rates, 16-QAM achieves a higher NGMI than PCS-16-QAM; however, this trend reverses for higher data rates. At the target net data rate of 400 Gbit/s, 16-QAM (at 60 GBd) achieves an average NGMI of 0.86, which is below the FEC threshold, whereas PCS-16-QAM (at 64 GBd) reaches a higher NGMI of 0.89, exceeding the FEC threshold. Representative received constellation diagrams at 64 GBd are shown in the inset of Fig. 3(a). To further validate realistic operation at this net data rate, we successfully decoded the best-performing configuration, i.e., PCS-16-QAM at 64 GBd, using an SC-LDPC code with rate 0.84 (corresponding to 19% FEC overhead) [17]. Accounting also for an additional 0.95% rate loss introduced by the distribution matcher, this achieves error-free transmission at a net rate of 400 Gbit/s. To investigate the difference between PCS-16-QAM and 16-QAM, Fig. 3(b) and (c) show the per polarization

SNR, directly computed from the received constellation after DSP, for both modulation formats. First, a small polarization-dependent imbalance of less than 0.5 dB is observed, attributed to residual imperfections in the optical modulator, in the UTC-PDs, and in the mixers of the receivers. At higher symbol rates, PCS-16-QAM offers a modest effective SNR improvement (0.6 dB at 64 GBd) reflecting its improved robustness in signal transmission at high spectral efficiencies. Combined with the theoretical NGMI gains of PCS [15], this explains the slightly higher NGMI shown in Fig. 3(a). In Fig. 3(d) the NGMI of PCS-16-QAM at 64 GBd is plotted as a function of the optical power measured just before the PBS. System performance improves monotonically up to an optimum of +19 dBm, beyond which it degrades due to nonlinear distortions in the UTC-PDs and, in particular, in the receiver front-end. The performance difference between the two polarizations remains largely independent of the transmit power. At the optimum operating point, the NGMI for each individual polarization exceeds the FEC threshold, further indicating that reliable decoding is achievable even with independent FEC applied per polarization. To visualize the channel characteristics, Fig. 3(e) displays the magnitude of 2×2 MIMO equalizer taps at 64 GBd with unitary polarization mixing in the optical domain.

Conclusion

This work reports a 400 Gbit/s net data rate over a 56-m single-aperture and single-channel dual-polarization sub-THz wireless link at a carrier frequency of 287 GHz, enabled by a novel ultra-broadband all-electronic mHEMT receiver, using 64-GBd PCS-16-QAM with SC-LDPC FEC, fully compliant with IEEE 802.15.3d. By demonstrating fiber-class throughput using realistic coding and digital signal processing in a single-channel architecture, this work establishes sub-THz point-to-point systems as a promising solution for next-generation, high-capacity wireless interconnects.

Acknowledgment

This work is financially supported by the German Federal Ministry of Research, Technology and Space (BMFTR) within the projects *Open6GHub* (16KISK010) and *Open6GHub+* (16KIS2405) and by the Dept. of Electronics and Telecommunications (DET) of Politecnico di Torino (*Incentivazione Dipartimentale – Azione A1.2*).

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